

JEE MAIN REVISION MASTER: ALCOHOL, PHENOLS AND ETHERS 2026 4 QUESTIONS 6.67% WEIGHTAGE

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 32 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Acidity of Phenols & Distinguishing Tests	16 PYQs	Phenols
Very High	Williamson Ether Synthesis & Cleavage by HI	16 PYQs	Ethers

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Acidity of Phenols & Distinguishing Tests	$\text{Acidity: Picric acid} > \text{p-Nitrophenol} > \text{Phenol} > \text{Water} > \text{Ethanol}$, $\text{Lucas: } 3^\circ \text{ (turbidity immediately)}, 2^\circ \text{ (5 mins)}, 1^\circ \text{ (no turbidity)}$
Williamson Ether Synthesis & Cleavage by HI	$\text{Williamson: } \text{R-O}^- + \text{Na}^+ + \text{R'-X (must be } 1^\circ) \rightarrow \text{R-O-R'} + \text{NaX}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Acidity of Phenols & Distinguishing Tests (Very High Yield — 16 PYQs)

Core Principle: Phenoxide resonance stabilization, substituent effects on acidity, Reimer-Tiemann and Kolbe-Schmitt reactions, and Lucas test.

■ **Caution / JEE Pitfall:** Electron withdrawing groups (-NO₂, -CN) INCREASE phenol acidity (especially at ortho and para positions), while electron donating groups (-CH₃) decrease acidity.

■ **Speed Trick:** Reimer-Tiemann reaction involves dichlorocarbene (:CCl₂) as an electrophilic intermediate to convert phenol to salicylaldehyde.

Representative Source PYQs:

• [2026 Shift] Q#2 (MCQ): "Glycerol is a trihydric alcohol. It contains ____." [Ans: B]

• [2026 Shift] Q#4 (MCQ):

$\text{C}_3\text{H}_6\text{O} \xrightarrow[\text{NaOH}]{\text{BH}_3}$ $\text{C}_3\text{H}_5\text{O} \xrightarrow{\text{CrO}_3}$... [Ans: N/A]

• [2025 Shift] Q#2 (MCQ): "Phenol can be distinguished from propanol by using the reagent..." [Ans: N/A]

3.2 Williamson Ether Synthesis & Cleavage by HI (Very High Yield — 16 PYQs)

Core Principle: Alkoxide S_N2 displacement on primary halides, and ether cleavage mechanism with concentrated HI.

■ **Caution / JEE Pitfall:** In Williamson synthesis, the alkyl halide **MUST** be primary (1°). If a tertiary (3°) halide is used, elimination dominates to give an **ALKENE** instead of an ether!

■ **Speed Trick:** With unsymmetrical ethers + HI: If one alkyl group is tertiary (3°), the iodide attaches to the 3° carbon (via S_N1). If both are 1° or 2° , iodide attacks the less hindered carbon (via S_N2).

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "The correct IUPAC name of is..." [Ans: N/A]
- [2026 Shift] Q#3 (MCQ): " $R-CH_2OH$ is converted into $R-CHO$ by reacting with _____." [Ans: N/A]
- [2025 Shift] Q#1 (MCQ): "Ethyl alcohol is heated with concentrated sulphuric acid at 413 K. The major product..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

The correct IUPAC name of is

- (A) Tertiary butoxy ethane
- (B) 1,1-Dimethyl-1-ethoxyethane
- (C) 2-ethoxy-2-methyl propane
- (D) Ethoxy tertiary butane

Correct Answer: None

Solution Analysis: In an ether, the larger carbon part is chosen as the parent hydrocarbon, and the smaller part attached through oxygen is named as an alkoxy group. Here, one side of oxygen is an ethyl group, $-C_2H_5$. So this part becomes the alkoxy group: ethoxy, $-OC_2H_5$. The other side is a tert-butyl group, $-C(CH_3)_3$. The longest continuous chain in this part has 3 carbon atoms, so the parent hydrocarbon is propane. In the propane chain, the middle carbon is carbon number 2. Both substituents are attached to this carbon. At carbon 2, there is one methyl group, $-CH_3$, and one etho

Benchmark Problem 2: KCET 2026 (Q#2)

Glycerol is a trihydric alcohol. It contains ____.

- (A) One primary, one secondary and one tertiary alcoholic groups
- (B) Two primary and one secondary alcoholic groups **[Correct]**
- (C) Two secondary and one primary alcoholic groups
- (D) One primary and two tertiary alcoholic groups

Correct Answer: B

Solution Analysis: Glycerol is a trihydric alcohol, so it contains three alcoholic groups. Its structure is:

$CH_2OH-CHOH-CH_2OH$ Now look at the carbon atoms attached to the $-OH$ groups. The two end carbon atoms are CH_2OH groups. Each of these carbon atoms is attached to only one other carbon atom, so these are primary alcoholic groups. The middle carbon atom is $CHOH$. It is attached to two other carbon atoms, so this is a secondary alcoholic group. So glycerol contains: two primary and one secondary alcoholic groups Correct option: B.